

Course milestone M8: Robustness Audit

HONR 46400 · Evidence-Driven Research

What to submit

One file, `lastname_m08.pdf`. Everything this milestone asks for goes inside it, as sections in the order below.

Submit it on Brightspace, under **Assignments** then **M08**. Brightspace carries the deadline.

Your work lives in a Colab notebook, which has no export-to-PDF button. **Making the PDF you hand in**, at the end of this document, is the one-minute routine that turns the notebook into the file you upload.

This milestone presents **Book Milestone 8: Your robustness audit** (checkpoint, version 1) of the course book, EDR|AI.

What this course adds

Your second mentor meeting, requested

The second of the course's two mentor meetings is booked now, well ahead of the poster work it exists to serve. Email your instructor again, from your Purdue address, asking for a meeting inside the Round 02 window your course platform posts.

The email carries the same three things the first request carried: at least three concrete slots inside the window, your project in two sentences as it now stands, and the one decision you most want pressure-tested. Add one line this time naming what has changed since Round 01. Save the sent email as a PDF and carry it as a section of your milestone document.

What Round 02 is for. By then your result exists and has been audited, and the poster is the next thing you owe the conference. So the meeting works on the claim your evidence can carry, the board that will carry it, and the questions a visitor will ask you in front of it.

Send this request early. The Round 02 window sits inside the busiest stretch of the conference block, and a request that arrives late gets you no meeting at all.

Milestone 8: Your robustness audit

Open the milestone workbook in Colab

This chapter closes Studio 8: Stress-test and adjudicate. Its lessons each ended at **It is your turn**; what you wrote there arrives here as working material, and this is where the pieces become one artifact you can defend.

What this milestone produces. A pre-listed robustness grid, negative tests with their assumptions stated, diagnostics, an adversarial-review record, and your adjudication of what survived.

What you bring

You also carry forward the last milestone's artifact: **Milestone 7: Your first reproducible analysis**. This one builds on it, and the chain assumes it is versioned and filed.

Check you are carrying each of these before you start. If one is missing, go back and work that lesson's **It is your turn** first; the practice below assumes it.

- Lesson 24: Robustness and Sensitivity: your pre-listed robustness grid with its full results, direction and size, plus the named null check.
- Lesson 25: Diagnostics and Negative Tests: your negative test and diagnostic, each with the prediction you wrote before running it.
- Lesson 26: AI as Adversarial Reviewer: your commissioned adversarial review with every flag adjudicated against the data, confirmed or refuted.

- Lesson 27: Recognizing False Confidence: your confidence audit: the not-yet-recomputed list, where belief in the headline number comes from, and your boundary sentence.

The practice

1. Pre-list your checks before you run any of them, and commit to reporting all of them.
2. Run the robustness grid and report a span within each commensurable panel, never across panels.
3. Run the null check that your design licenses, and name it correctly.
4. Commission at least one adversarial review, and add a human reviewer when one is available. Neither review is independent verification: confirm or refute every flag with a data check.
5. Write the adjudication: what survived, what did not, and what you still cannot rule out.

A version, not a pass

Your milestone artifact is a dated, numbered **version** with the reason for the version attached. When later evidence changes it, you write the next version rather than editing the last one, because the sequence of changes is itself part of your research record.

The four rails, here

Four concerns cross every studio in this book. At this milestone they take these specific forms.

- **Ethics, permissions, and data exposure.** Report the checks that hurt your finding as fully as the ones that helped.
- **Evidence, provenance, and reproducibility.** A flag is real when a check confirms it, not when the reviewer sounds certain.
- **AI activity, verification, and human decisions.** An AI reviewer is another critique, not independent verification.
- **Uncertainty, claim boundary, and revision history.** Specification spread measures your choices; it is not an uncertainty interval.

Where this milestone sits

You arrived carrying Milestone 7: Your first reproducible analysis. Milestone 9 writes down only the claims this audit licenses, at the strength it licenses them. The chain ends at Milestone 12: Your release and next cycle, where you decide whether your finished research artifact leaves your hands.

What sends you back

A finding that does not survive its own pre-listed checks returns to Studio 5 or 7, not to the writing studio.

How this milestone is assessed

Each row scores 0, 1, or 2. **10 points total.**

#	Criterion	0	1	2
1	The milestone artifact exists in full, specific to your own project	absent, or generic text that would fit any project	present but incomplete, or specific in some parts and generic in others	complete and unmistakably about your project
2	The milestone is written as a dated, numbered version with its reason	no version, or a version with no reason given	versioned, but the reason is decorative rather than an account of what changed	versioned with a reason a reader could use to reconstruct your thinking
3	Every judgment in the artifact is defended by you, not asserted by a tool	conclusions appear with no reasoning, or reproduce a tool's wording	reasoning present but thin where the decision was hardest	each judgment carries a reason you could defend under questioning
4	The four rails are carried in the form this studio requires	one or more rails absent where this studio required them	all rails present but one is nominal	every rail addressed in the form this studio requires

#	Criterion	0	1	2
5	The milestone's own product is present and defensible: A pre-listed robustness grid, negative tests with their assumptions stated, diagnostics, an adversarial-review record, and your adjudication of what survived	the milestone's own product is missing	present but would not survive a careful question	present and defensible as stated

Your workbook

Open the workbook with the link above. It walks these steps with a cell for each one, ends by writing your milestone version with its reason, and adds the rows this studio contributes to your AI Research Ledger and your Research Dossier.

Making the PDF you hand in

Colab has no export-to-PDF button. You make the PDF by **printing the notebook to a file**, which takes about a minute once you know the preparation steps. Skipping them is the most common way a milestone arrives looking half finished, because a notebook prints what is on the screen and nothing else.

1. Prepare the notebook

1. **Run the whole thing.** `Runtime` → `Run all`, then wait for it to finish. A cell you never ran prints with nothing underneath it, so an analysis you really did can reach the grader as an empty box.
2. **Expand every collapsed section.** Anything folded shut prints as a heading with nothing beneath it. Open all of them, including any Colab folded for you.
3. **Shorten any output that scrolls.** When Colab puts a scrollbar on an output, or tells you the output is truncated, only the visible part prints. Show the first twenty rows instead of the whole table, or summarise it, then rerun that cell. A reader did not want the whole table anyway.
4. **Read it once, top to bottom.** You are about to freeze it.

2. Print it to a file

`File` → `Print` opens your browser's print dialog. Set the destination to **Save as PDF**, open the dialog's further settings, and turn on **Background graphics**, so code cells keep their shading and your figures keep their backgrounds. Save it under the file name this milestone asks for.

Wide tables and wide figures are the usual casualty. If something runs off the right edge, switch the layout to **landscape** or reduce the scale, and print again.

3. Check the PDF before you upload it

Open the file you just made and confirm three things: every figure is whole rather than sliced by a page break, no output stops mid-line, and the pages run in order with nothing missing. A PDF is an artifact like any other in this course, so you verify it before your name goes on it.

One route not to take

Ask an AI assistant how to export a Colab notebook and it will very likely hand you `jupyter nbconvert --to pdf`. Do not spend your afternoon on it. Inside Colab that route needs a large LaTeX installation, runs for several minutes, and then either fails or quietly drops the symbols and emoji these notebooks are full of. Printing to a file is the route that works.

That is a small, cheap instance of the rule this whole course runs on. The confident answer is not automatically the correct one, and the way you find out is to check it against what actually happens.